```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
```

```
from google.colab import files
uploaded=files.upload()
```

Choose files No file chosen Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

```
Saving titanic .csv to titanic  (2).csv
```

```
df = pd.read_csv("titanic .csv")
```

```
df
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	C
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	
...	...	...	...	...	...	...	...	...	...	...	
886	887	0	2	Montvila, Rev. Juozas	male	27.0	0	0	211536	13.0000	
887	888	1	1	Graham, Miss. Margaret Edith	female	19.0	0	0	112053	30.0000	
888	889	0	3	Johnston, Miss. Catherine Helen "Carrie"	female	NaN	1	2	W./C. 6607	23.4500	
889	890	1	1	Behr, Mr. Karl Howell	male	26.0	0	0	111369	30.0000	
890	891	0	3	Dooley, Mr. Patrick	male	32.0	0	0	370376	7.7500	

891 rows × 12 columns

```
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cab
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	N
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	N
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C1
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	N

```
df.shape
```

```
(891, 12)
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

```
df.duplicated().value_counts()
```

	count
False	891

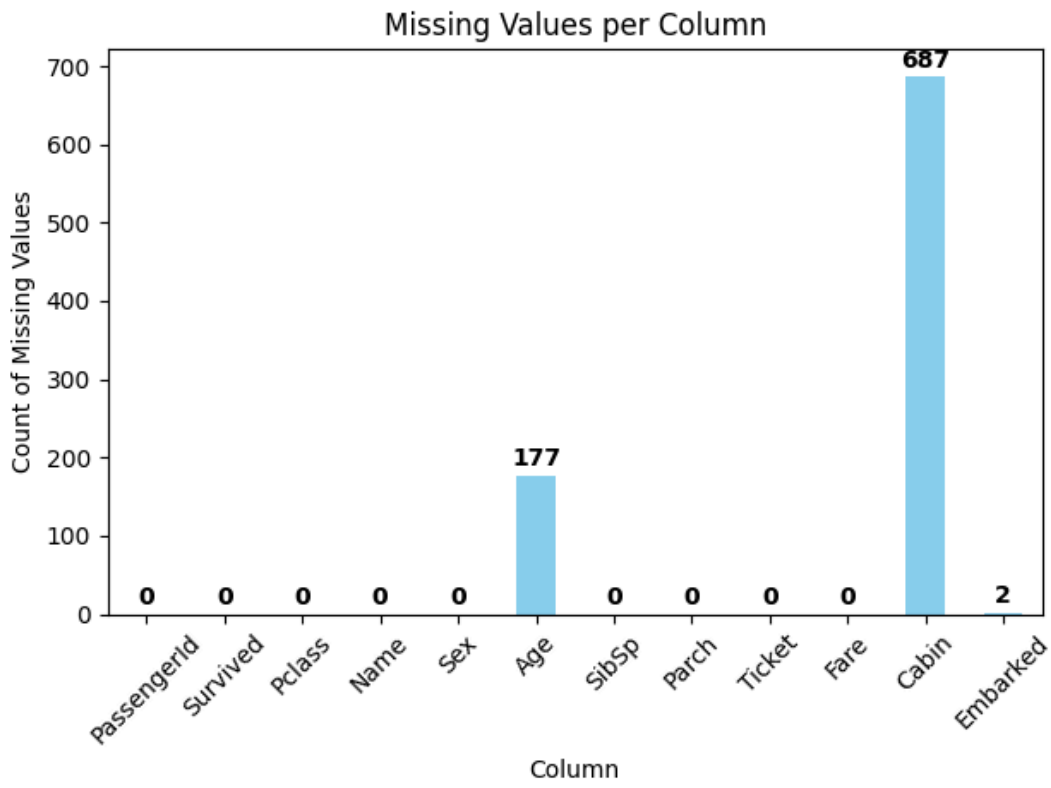
dtype: int64

```
# Check how many missing values exist in each column
print(df.isnull().sum())
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

```
missing = df.isnull().sum()
```

```
# Create the bar plot
ax = missing.plot(kind='bar', color='skyblue')
plt.title('Missing Values per Column')
plt.ylabel('Count of Missing Values')
plt.xlabel('Column')
plt.xticks(rotation=45)
#  Add text labels above each bar
for index, value in enumerate(missing):
    plt.text(index, value+5, str(value), ha='center', va='bottom', fontsize=10, fontwei
plt.tight_layout()
plt.show()
```




```
df = df.drop(columns=['Cabin']) # Cabin has 687 missing — too many!
```

```
df['Age'] = df['Age'].fillna(df['Age'].median())
```

```
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

```
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Embarked	0

dtype: int64

```
# Count the number of people who survived and didn't
survival_counts = df['Survived'].value_counts()
print("Survival Counts:\n", survival_counts)
```

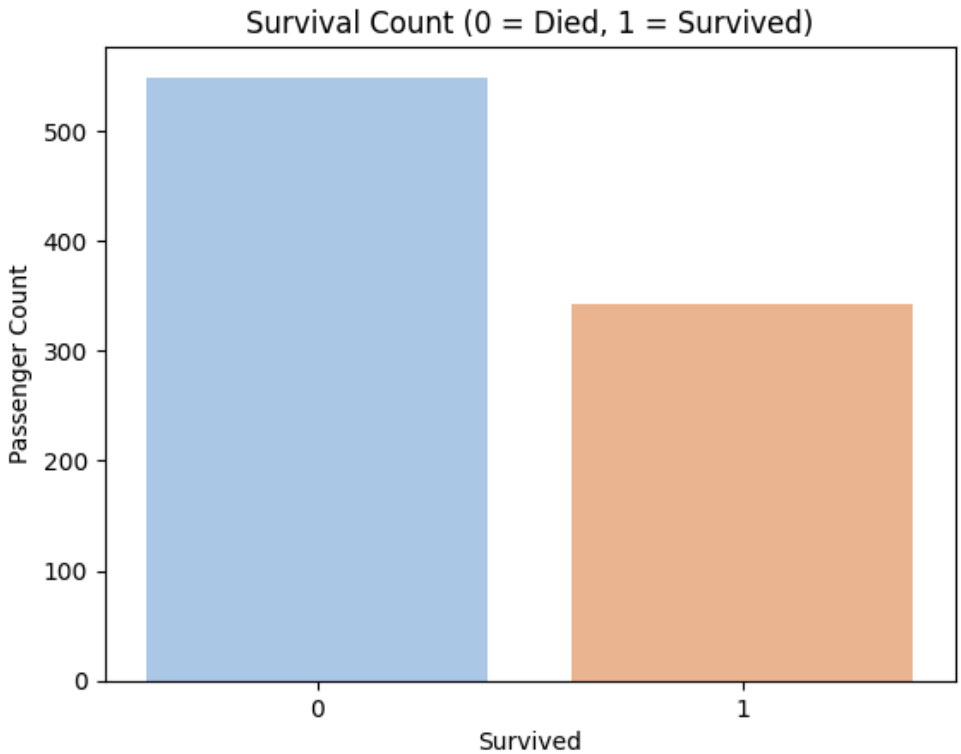
```
Survival Counts:
 Survived
0    549
1    342
Name: count, dtype: int64
```

```
#Survival Count (Bar Plot)
sns.countplot(x='Survived', data=df, palette='pastel')
plt.title('Survival Count (0 = Died, 1 = Survived)')
plt.xlabel('Survived')
plt.ylabel('Passenger Count')
plt.show()
```

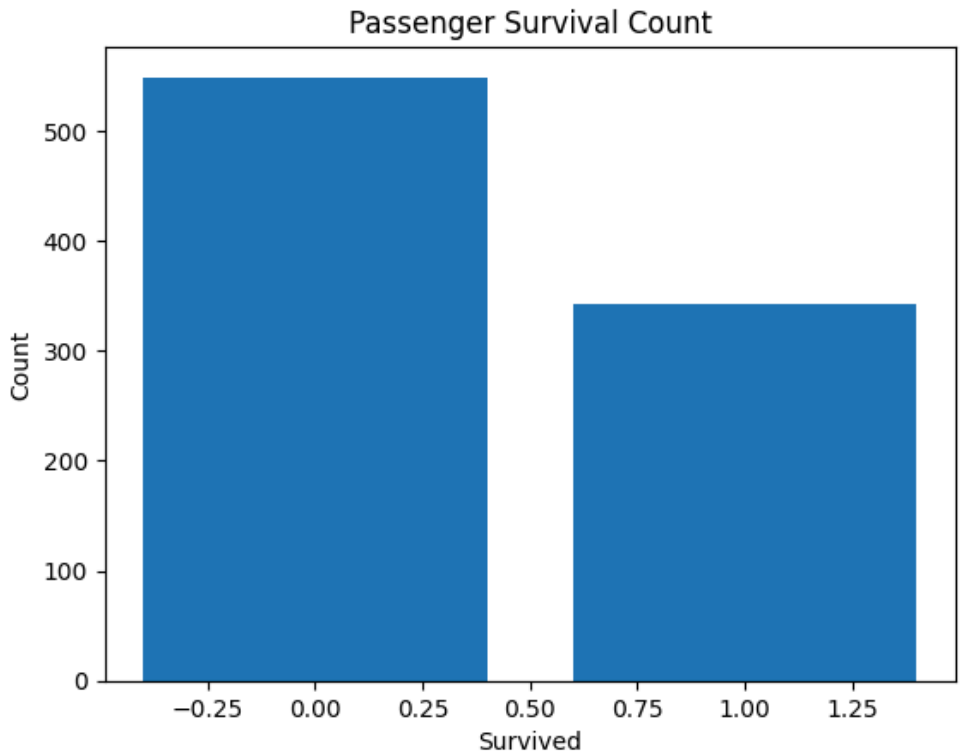
```
/tmp/ipykernel_4163/927445151.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0.

  sns.countplot(x='Survived', data=df, palette='pastel')
```



```
counts = df['Survived'].value_counts() # counts of 0 and 1
plt.bar(counts.index, counts.values)
plt.xlabel("Survived")
plt.ylabel("Count")
plt.title("Passenger Survival Count")
plt.show()
```




```
# Count of passengers by gender
gender_counts = df['Sex'].value_counts()
print("\nGender Counts:\n", gender_counts)
```

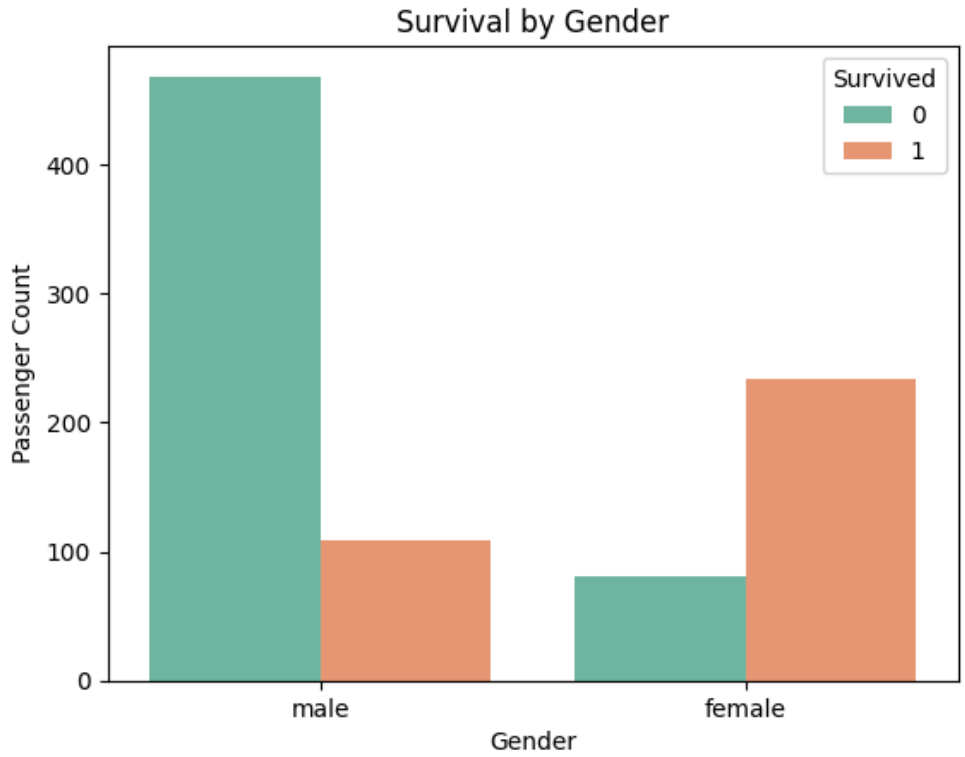
```
Gender Counts:
 Sex
male      577
female    314
Name: count, dtype: int64
```

```
# Count of survived grouped by gender
survival_by_gender = df.groupby(['Sex', 'Survived']).size()
print("\nSurvival by Gender:\n", survival_by_gender)
```

```
Survival by Gender:
 Sex     Survived
female  0            81
        1           233
male    0           468
        1           109
dtype: int64
```

```
#Survival by Gender
sns.countplot(x='Sex', hue='Survived', data=df, palette='Set2')
```

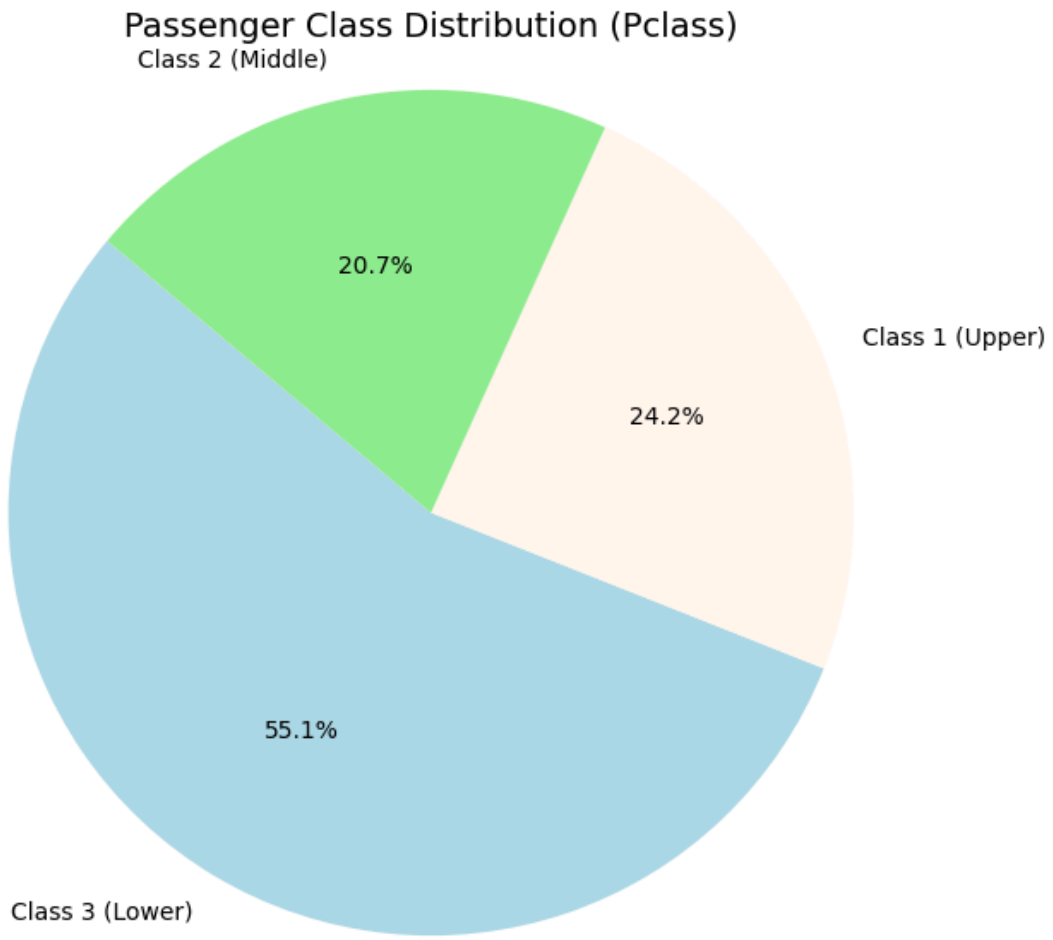
```
plt.title('Survival by Gender')
plt.xlabel('Gender')
plt.ylabel('Passenger Count')
plt.legend(title='Survived')
plt.show()
```




```
#Passenger Class Distribution
# Count of passengers by class
pclass_counts = df['Pclass'].value_counts()
print("\nPassenger Class Counts:\n", pclass_counts)
```

```
Passenger Class Counts:
 Pclass
3    491
1    216
2    184
Name: count, dtype: int64
```

```
#Passenger Class Distribution
# Labels for the pie chart
labels = ['Class 3 (Lower)', 'Class 1 (Upper)', 'Class 2 (Middle)']
# Plotting the pie chart
plt.figure(figsize=(7, 7))
plt.pie(pclass_counts, labels=labels, autopct='%1.1f%%', startangle=140, colors=['light
plt.title('Passenger Class Distribution (Pclass)', fontsize=14)
plt.axis('equal') # Ensures it's a circle
plt.show()
```




```
#Age Distribution ( KDE Plot (Smooth Curve Distribution
sns.kdeplot(df['Age'], fill=True, color='skyblue')
plt.title('Age Distribution (KDE Plot)')
plt.xlabel('Age')
plt.ylabel('Density')
plt.show()
```

Age Distribution (KDE Plot)